Low-Light Wildlife Detection with Event Cameras

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Abstract

We detect nocturnal animals with an event camera and a spiking neural detector.

At 2 lux the detector reaches a mean recall of 0.71 across nine species.

Introduction

Camera traps at night rely on infrared illumination, which can disturb some animals.

Event cameras report brightness changes per pixel and work at very low light levels.

Their sparse output suits spiking detectors that run on low-power hardware in the field.

Few studies measure how detection quality changes as the scene gets darker.

We collect clips in forests and report recall at several controlled light levels.

Our aim is to find the light level below which detection becomes unreliable.

We also compare the power draw of the event camera with an infrared reference camera.

The dataset and labels are described in the methods section of this paper.

Methods

Events are accumulated into 4 temporal bins of 25 milliseconds each.

The detector was trained on 14,200 labelled clips from 3 forest sites.

Clips were labelled by two ecologists who marked the species and a bounding box.

Light levels were measured with a calibrated lux meter placed next to the camera.

We report recall because false negatives matter most for population surveys.

Each site contributed clips from both summer and winter recording periods.

Training used a fixed learning schedule without any per-site tuning at all.

Test clips come from recording nights that were not used during training.

Results

Figure 3. Recall per species at 2 lux.

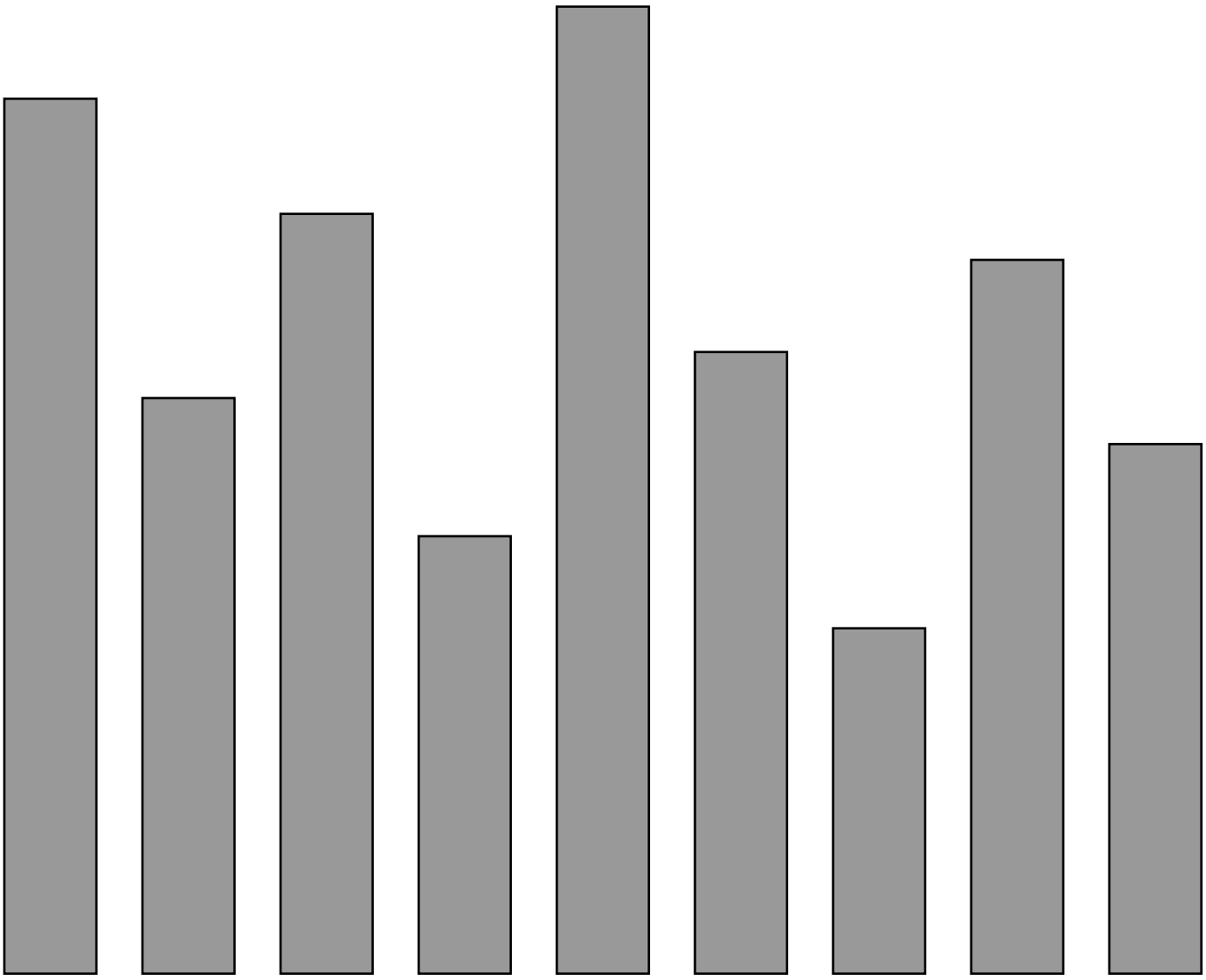

Figure 3 shows per-species recall; the exact values appear only in the figure.
Mean recall fell from 0.71 at 2 lux to 0.52 at 0.5 lux across all nine species.
False positives were most often caused by moving branches in windy conditions.
Recall was stable across the two recording seasons for most of the species.
Small animals such as mice were detected less often than larger mammals.
Detection latency on the field hardware stayed below 40 milliseconds per clip.
Adding a fifth temporal bin did not change mean recall in our ablation runs.
Clips with rain produced more events but did not reduce recall noticeably.

Discussion

The event camera consumed 0.9 W compared with 2.6 W for the infrared reference camera.
Lower power allows longer deployments on the same battery in remote locations.
Detection below 0.5 lux may need longer bins or a more sensitive sensor model.
Branch motion remains the main source of errors and needs better filtering.
Our sites were temperate forests, so other habitats may behave differently.
Species with little motion at night are hard to detect with events alone.
A combined event and infrared system could cover both situations in future.
Ethics approval covered passive recording only, without any animal handling.

Conclusion

Event cameras detect nocturnal wildlife reliably down to about 2 lux at low power. Per-species values should be read from Figure 3 in the original file.

References

[1] M. Haddad. Spiking Detectors for Neuromorphic Vision. 2024.

[2] B. Keller. Camera Traps and Animal Disturbance. 2022.